

Mathematical Review: RealityOfZeros

April 23, 2026

Abstract

This is a mathematical review of the document *Band-Limitedness and Reality of Zeros of the Inverse-Phase Pullback of the Hardy Z-Function* (file `RealityOfZeros.tex`, dated April 23, 2026), conducted according to the math-referee protocol. Auxiliary sources consulted: `BandLimitedness.tex` (establishes $\text{supp } \rho \subseteq [-1, 1]$) and `three_missing_theorems.pdf` (Theorem B, diagonal-dominance identity for the whitened differential). The review covers mathematical content only. Every finding is classified as **Fixable** or **Not fixable** at the start of the finding.

Findings

Finding 1 — Fixable

Location: Lemma (HB reduction), proof, sentence beginning “bounded by 1 on $\mathbb{R}$... hence by the maximum principle $|Y_+^*(z)/Y_+(z)| \leq 1$ for $z \in \mathbb{C}^+$ ”.

Claim as written: If Y_+ has no zeros in $\mathbb{C}^+$, then Y_+^*/Y_+ is holomorphic in $\mathbb{C}^+$, bounded by 1 on $\mathbb{R}$, hence $|Y_+^*(z)/Y_+(z)| \leq 1$ for $z \in \mathbb{C}^+$ by the maximum principle.

What is relied on but not established in the text: The maximum-principle step requires not only that $|Y_+^*/Y_+| \leq 1$ on $\mathbb{R}$ but also that Y_+^*/Y_+ is *bounded* in $\mathbb{C}^+$ (i.e. belongs to $H^\infty(\mathbb{C}^+)$) or, alternatively, that it satisfies the hypotheses of the Phragmén–Lindelöf principle on $\mathbb{C}^+$. The text asserts that Y_+ has mean type 0 in $\mathbb{C}^+$, so Y_+ is bounded in $\mathbb{C}^+$, but Y_+^* has mean type 1 in $\mathbb{C}^+$ (it grows like $e^{|y|}$ along $i\mathbb{R}^+$, as noted explicitly in Step 2 of the Main Structural Theorem). Consequently, Y_+^*/Y_+ is *not* bounded in $\mathbb{C}^+$; it grows at most like $e^{|y|}$ in $\mathbb{C}^+$. The standard maximum principle (for bounded functions) therefore does not apply directly. What is needed is an explicit invocation of the Phragmén–Lindelöf principle for the half-plane: if a function f holomorphic in $\mathbb{C}^+$ satisfies $|f(z)| \leq 1$ on $\mathbb{R}$ and $|f(z)| = O(e^{\tau|z|})$ in $\mathbb{C}^+$ for some $\tau < \infty$, then $|f(z)| \leq 1$ throughout $\mathbb{C}^+$. The function Y_+^*/Y_+ satisfies the growth bound $|Y_+^*(z)/Y_+(z)| \leq Ce^y$ for $z = x + iy \in \mathbb{C}^+$, and this is exactly the form required by the standard Phragmén–Lindelöf theorem on the upper half-plane with exponent $\tau = 1 < \pi/(\pi) = 1$ (i.e. at the boundary of applicability). The correct theorem is the Phragmén–Lindelöf theorem for a half-plane with growth of order 1: a function holomorphic on $\mathbb{C}^+$, continuous on $\overline{\mathbb{C}^+}$, bounded by 1 on $\mathbb{R}$, and satisfying $\log |f(x + iy)| = O(y)$ as $y \rightarrow +\infty$, is bounded by 1 on all of $\mathbb{C}^+$. That this exponent is exactly order 1 means the classical form applies without additional logarithmic corrections, but the hypothesis $\log |f(z)| = O(|z|)$ must be verified (not just growth of order at most 1) and must be stated.

What would establish it: Replace “by the maximum principle” with an explicit invocation of the Phragmén–Lindelöf theorem for the half-plane (e.g. Titchmarsh, *The Theory of Functions*,

Theorem 5.63): state that Y_+^*/Y_+ is holomorphic in $\mathbb{C}^+$, continuous on $\overline{\mathbb{C}^+}$ (which follows from Y_+ having no zeros in $\mathbb{C}^+$ and Y_+ being bounded there), satisfies $|Y_+^*(z)/Y_+(z)| \leq 1$ on $\mathbb{R}$, and satisfies $\log |Y_+^*(z)/Y_+(z)| = O(\operatorname{Im} z)$ for $z \in \mathbb{C}^+$ (which follows from the indicator bound: Y_+^* has mean type 1 in $\mathbb{C}^+$ and Y_+ is bounded away from zero on $\{|z| > R\} \cap \mathbb{C}^+$ for some R because Y_+ is outer in $\mathbb{C}^+$ and the Szegő condition holds, so $|Y_+(iy)| \not\rightarrow 0$ faster than any exponential in y). Once these hypotheses are in place, the Phragmén–Lindelöf conclusion $|Y_+^*(z)/Y_+(z)| \leq 1$ for $z \in \mathbb{C}^+$ follows.

Finding 2 — Fixable

Location: Main Structural Theorem, Step 1, sentence “Since B and O have nonnegative mean type in $\mathbb{C}^+$, and $S(iy) = e^{-\alpha y}$ contributes mean type $-\alpha \leq 0$, the only way for the total mean type to be 0 is $\alpha = 0$ ”.

Claim as written: The mean types of B , $S = e^{i\alpha z}$ ($\alpha \geq 0$), and O in $\mathbb{C}^+$ sum to the mean type of Y_+ in $\mathbb{C}^+$, which equals 0; since B and O have nonnegative mean type, and S has mean type $-\alpha \leq 0$, we must have $\alpha = 0$.

What is relied on but not established in the text: The argument requires the claim that Blaschke products B and outer functions O in $H^\infty(\mathbb{C}^+)$ have nonnegative mean type in $\mathbb{C}^+$. For the Blaschke product this is standard (Blaschke products have mean type 0 in $\mathbb{C}^+$ under the Blaschke convergence condition, by Jensen’s formula applied to the half-plane), and the document states “Blaschke products have zero mean type” in Step 2. For the outer factor O , the claim that outer functions in $H^\infty(\mathbb{C}^+)$ have nonnegative mean type in $\mathbb{C}^+$ is not a consequence of the definition of “outer” alone: the mean type of an outer function can in principle be any real number. The statement that is true and sufficient is: for an entire function of exponential type with one-sided Fourier support $[0, \sigma]$, the mean type in $\mathbb{C}^+$ equals $-\inf \operatorname{supp} \hat{f} = 0$, and the mean type in $\mathbb{C}^-$ equals $\sup \operatorname{supp} \hat{f} = \sigma$ (this is the content of the Krein–Akhiezer theorem cited in Step 3, but it is not invoked in Step 1). In Step 1 the document has not yet established the Fourier-support identification of Step 3; it is therefore relying on a fact (nonnegativity of the mean type of O in $\mathbb{C}^+$) that is only justified later in the proof, creating a logical dependency that is not acknowledged.

What would establish it: Reorder the argument so that the mean-type constraint in Step 1 is applied *after* Step 3 establishes the Fourier support; or cite explicitly, already in Step 1, that the inner-outer factorization of an $H^\infty(\mathbb{C}^+)$ function that extends to an entire function of exponential type with indicator $h(\pi/2) = 0$ forces $\operatorname{meantype}(O) \geq 0$ in $\mathbb{C}^+$ by the Paley–Wiener theorem applied to O directly: since O is outer and $|O| = |Y_+|$ on $\mathbb{R}$ with Y_+ bounded on $\mathbb{R}$, $O \in H^2(\mathbb{C}^+) \cap H^\infty(\mathbb{C}^+)$, and the indicator $h_O(\pi/2) \leq h_{Y_+}(\pi/2) = 0$ gives the mean type in $\mathbb{C}^+$ equal to 0 (more precisely: mean type $= h(\pi/2)$ for functions of exponential type, by definition, and $h_O(\pi/2) \leq h_{Y_+}(\pi/2) = 0$ by the inner-outer factorization and $|B| \leq 1$ on $i\mathbb{R}^+$).

Finding 3 — Fixable

Location: Main Structural Theorem, Step 3, sentence “the outer factor of Y_+ has the same Fourier support $[0, 1]$ ”.

Claim as written: In the Cartwright–Levinson representation $Y_+ = Ce^{iaz}B(z)O(z)$, the outer factor O has Fourier support $[0, 1]$.

What is relied on but not established in the text: The document asserts that the Fourier support of O is determined by the Szegő construction from the boundary modulus $|O| = |Y_+|$ on $\mathbb{R}$.

This appeal to “Akhiezer’s theorem” (that the outer factor is the unique outer function with given boundary modulus, given the Szegő condition) correctly identifies O as the unique outer function with $|O(u)| = |Y_+(u)|$ on $\mathbb{R}$. However, the Fourier support of this outer function is $[0, 1]$ only if the Fourier transform of $\log |Y_+|$ is supported in $[0, 1]$ in the appropriate sense; more precisely, the Fourier support of an outer function $O \in H^2(\mathbb{C}^+)$ is determined by its argument on $\mathbb{R}$ (via the Hilbert-transform relation between $\log |O|$ and $\arg O$ on $\mathbb{R}$), not by the support of $\log |O|$ alone. Knowing $|O(u)| = |Y_+(u)|$ on $\mathbb{R}$ determines $\log |O|$ on $\mathbb{R}$ but does not by itself force the Fourier support of O (as a function) to equal $[0, 1]$: O could have Fourier support $[-a, b]$ for various $a, b \geq 0$ with $a + b = 1$, depending on the argument of O on $\mathbb{R}$. The correct fact that would close this gap is the combination of: (a) O is outer in $H^2(\mathbb{C}^+)$ with boundary value in $L^2(\mathbb{R})$; (b) O has exponential type ≤ 1 (inherited from Y_+); (c) O is bounded on $\overline{\mathbb{C}^+}$ (as an H^∞ outer factor); (d) O has mean type 0 in $\mathbb{C}^+$ and mean type 1 in $\mathbb{C}^-$ (established by the indicator argument in Step 3, since $a = 0$ and $\tau(O) = 1$). Items (b)–(d) together, by the Paley–Wiener theorem for functions of exponential type with indicator, force $\text{supp } \hat{O} \subseteq [0, 1]$. Showing $\text{supp } \hat{O} = [0, 1]$ (not a proper subset) requires that the spectral measure of O charges a neighborhood of both endpoints 0 and 1, which follows from $|O| = |Y_+|$ on $\mathbb{R}$ and the fact that Y_+ has full support in $[0, 1]$; this last fact is stated in the document but not proved.

What would establish it: Invoke the Paley–Wiener theorem for entire functions of exponential type with given indicator: since $h_O(\pi/2) = 0$ and $h_O(-\pi/2) = 1$ (i.e., mean type 0 in $\mathbb{C}^+$ and 1 in $\mathbb{C}^-$), any entire function O of exponential type satisfying these indicator values has Fourier support contained in $[0, 1]$ (the lower bound $-h_O(\pi/2) = 0$ and the upper bound $h_O(-\pi/2) = 1$ are exactly the Fourier-support endpoints by the Bohr–Carathéodory theorem for measures, see e.g. Levin, *Lectures on Entire Functions*, Ch. II). This argument is self-contained and does not require knowledge of $|O|$ beyond what is already available.

Finding 4 — Fixable

Location: Main Structural Theorem, Step 4, sentence beginning “Lower bound: $|Y_+(u)| \geq |Y(u)|$, and $|Y(u)| \geq$ the amplitude of the dominant Riemann–Siegel mode. . . which is ≥ 2 . . . So $\log |Y_+(u)| \geq 0$ on the set where $|Y(u)| \geq 1$, which has positive density in $\mathbb{R}$ by the equidistribution of $\vartheta(t)$ modulo 2π ”.

Claim as written: The set $\{u : |Y(u)| \geq 1\}$ has positive density in $\mathbb{R}$ because $\vartheta(t)$ equidistributes modulo 2π .

- What is relied on but not established in the text:**
1. The claim that the dominant Riemann–Siegel mode ($n = 1$) has amplitude ≥ 2 is a statement about $Y(u)$, not directly about $Z(t)$: the text writes $|Y(u)| \geq 2$ arising from “the $n = 1$ mode has amplitude 2”. More precisely, after the warp, the $n = 1$ mode of Y at $u = \Theta(t)$ is $2 \cos(\vartheta(t) - 0) / \sqrt{\Theta'(t)} \cdot \sqrt{\Theta'(t)} = 2 \cos(\vartheta(t))$, so the mode amplitude is $|2 \cos(\vartheta(t))|$, which oscillates between 0 and 2 rather than being uniformly ≥ 2 . The claim “amplitude ≥ 2 ” is therefore imprecise as written; the amplitude equals $2|\cos(\vartheta(t))|$, which attains arbitrarily small values near $\vartheta(t) = \pi/2 + k\pi$.
 2. The correct lower bound relevant to the Szegő integral is not that $|Y_+(u)| \geq 2$ pointwise on a set of positive measure, but that $\int_{\mathbb{R}} \frac{\log |Y_+(u)|}{1+u^2} du > -\infty$, which requires controlling the *integral* of $\log |Y_+(u)|$ rather than pointwise lower bounds. The argument as written uses a positive-density lower bound to conclude global integrability, but this step requires: (a) a quantitative lower bound on the measure of the set $\{|Y_+(u)| \geq c\}$ for some $c > 0$, which requires more than the qualitative equidistribution claim; (b) an upper bound on $\log^- |Y_+(u)| = \max(-\log |Y_+(u)|, 0)$

valid near simultaneous near-zeros of Y and $\mathcal{H}[Y]$, which is asserted by “simultaneous zeros of Y and $\mathcal{H}[Y]$ are a codimension-2 condition on $\mathbb{R}$, hence empty generically” but this heuristic requires a proof that the integral $\int \log^- |Y_+|/(1+u^2) du$ is finite.

3. The equidistribution of $\vartheta(t) \bmod 2\pi$ is invoked without reference. This is a standard consequence of Weyl’s theorem and the asymptotics $\vartheta(t) \sim \frac{t}{2} \log \frac{t}{2\pi}$ (which grows without bound and has derivative $\frac{1}{2} \log \frac{t}{2\pi} \rightarrow \infty$), so Weyl’s theorem applies and equidistribution holds. This step is routine but should be stated.

What would establish it: Replace the heuristic amplitude argument with a direct Szegő-condition proof: use the L^∞ upper bound $|Y_+(u)| \leq 2\|d\xi\|_{\text{TV}} < \infty$ (already noted in the document) to bound $\log^+ |Y_+|/(1+u^2)$ uniformly, so $\int \log^+ |Y_+|/(1+u^2) du < \infty$. For $\int \log^- |Y_+|/(1+u^2) du < \infty$: use Jensen’s inequality applied to the representation $Y_+(z) = 2 \int_0^1 e^{i\mu z} d\xi(\mu)$; for $z = iy$ with $y > 0$, $|Y_+(iy)| \geq 2|d\xi(\{0\})|$ if ξ has an atom at 0; the general case (absolutely continuous ρ) uses the Szegő theorem that O exists and is nonzero exactly when $\int \log \rho' d\mu > -\infty$ (with ρ' the Radon–Nikodym density of ρ), which is an independent condition on ρ that should be verified from the structure of the spectral measure of Y . Alternatively, one may invoke the fact that $Y_+ \in H^2(\mathbb{C}^+)$ with $|Y_+(u)|^2 = \rho'(u)$ (the spectral density), and verify that $\log \rho' \in L^1(\mathbb{R}, (1+u^2)^{-1} du)$ from the explicit form of ρ' .

Finding 5 — Fixable

Location: Main Structural Theorem, Step 5, first paragraph, sentence “the outer factor O is the unique entire outer function of exponential type 1 with that modulus, its Fourier support is determined by the modulus alone, hence equals $[0, 1]$ ”.

Claim as written: The Fourier support of the outer factor O is determined solely by its boundary modulus $|O| = |Y_+|$ on $\mathbb{R}$.

What is relied on but not established in the text: The Fourier support of an outer function $O \in H^2(\mathbb{C}^+)$ is *not* determined by its boundary modulus alone. Given $|O(u)| = f(u)$ for $u \in \mathbb{R}$, the outer function is uniquely determined (up to unimodular constant) by

$$\log O(z) = \frac{i}{\pi} \int_{\mathbb{R}} \frac{\log f(u)}{u - z} du, \quad z \in \mathbb{C}^+,$$

(the Herglotz/Poisson representation for the log), and the Fourier transform of O depends on both $\log f$ and the Hilbert transform of $\log f$ (which gives $\arg O$ on $\mathbb{R}$). Therefore the statement “its Fourier support is determined by the modulus alone” is not a consequence of the uniqueness of the outer function: uniqueness of O given $|O|$ on $\mathbb{R}$ is correct, but that uniqueness implies uniqueness of the Fourier support (rather than just its determination by the modulus via the Hilbert transform) is the claim that needs proof. What *is* true is that the indicator of O (and hence its Fourier support as an entire function of exponential type) is determined by $|O|$ on $\mathbb{R}$ via the Paley–Wiener theorem and the indicator formula, but this is a nontrivial statement that requires the hypothesis that O is entire of exponential type. The argument needed was already identified in Finding 3; the issue recurs here in the form of an unqualified assertion.

What would establish it: State explicitly that the determination of the Fourier support proceeds via the indicator, not via the modulus directly: since O is the unique outer function in $H^2(\mathbb{C}^+)$ with $|O| = |Y_+|$ on $\mathbb{R}$, and since O extends to an entire function of exponential type (because Y_+ does and $O = Y_+/B$ with B a Blaschke product that is entire of exponential type 0), the indicator

of O satisfies $h_O = h_{Y_+}$ (because $|B| = 1$ on $\mathbb{R}$ and the indicator of a product equals the sum of indicators for entire functions of exponential type with the same type), giving $\text{supp } \hat{O} = [0, 1]$ by the Paley–Wiener theorem, not from the modulus alone.

Finding 6 — Fixable

Location: Main Structural Theorem, Step 5, second paragraph, sentence “An entire Blaschke product on $\mathbb{C}^+$ that is a ratio of two entire functions of the same exponential type and same modulus on $\mathbb{R}$ must be a finite Blaschke product (by the Hadamard factorization of Y_+ and O : both have the same order and type, same modulus on $\mathbb{R}$, differing only by a finite set of zeros in $\mathbb{C}^+$)”.

Claim as written: An entire function that is simultaneously a Blaschke product on $\mathbb{C}^+$ and a ratio of two entire functions of the same exponential type and same boundary modulus on $\mathbb{R}$ must be a finite Blaschke product.

What is relied on but not established in the text: The parenthetical justification invokes Hadamard factorization and concludes that Y_+ and O “differ only by a finite set of zeros in $\mathbb{C}^+$ ”, which is exactly the claim that B is a finite Blaschke product — the very thing that needs to be proved. The argument is circular as written: it assumes the conclusion in the justification. What a non-circular argument requires is:

1. Both Y_+ and O are entire of exponential type 1, same order 1, same type 1 (by the indicator/Paley–Wiener argument).
2. They have the same modulus on $\mathbb{R}$, so $|Y_+(u)| = |O(u)|$ for all $u \in \mathbb{R}$.
3. Their ratio $B = Y_+/O$ is a Blaschke product on $\mathbb{C}^+$; as an entire function, B satisfies $|B(u)| = 1$ for all $u \in \mathbb{R}$ and is bounded by 1 on $\mathbb{C}^+$.
4. A non-constant infinite Blaschke product $\prod_n (z - z_n)/(z - \bar{z}_n)$ that is also entire must have its zeros $\{z_n\}$ in $\mathbb{C}^+$ satisfy the Blaschke condition $\sum_n \text{Im } z_n/(1 + |z_n|^2) < \infty$, but an entire function of exponential type τ has at most $O(r)$ zeros in a disk of radius r by Jensen’s formula, and the zeros of B in $\mathbb{C}^+$ inherit the zero-counting bound of Y_+ . The argument from Hadamard factorization that an *infinite* Blaschke product of exponential type 0 (since B has indicator 0 in all directions, as both $h_{Y_+} = h_O$ gives $h_B = 0$) has only finitely many zeros requires an additional step: an entire function of exponential type 0 that is also a Blaschke product bounded by 1 on $\mathbb{C}^+$ is *not* necessarily a finite Blaschke product from type considerations alone. The correct argument is that B is entire and of exponential type 0, with $|B(u)| = 1$ on $\mathbb{R}$ and $|B(z)| \leq 1$ on $\mathbb{C}^+$: by the Phragmén–Lindelöf principle applied to B and $1/B$ (the latter is meromorphic on $\mathbb{C}^+$ with poles only at zeros of B , but B is entire, so $1/B$ is meromorphic on all of $\mathbb{C}$), an entire function of exponential type 0 with $|f(u)| = 1$ on $\mathbb{R}$ must be a finite Blaschke product times a unimodular constant (this is a theorem in the theory of bounded analytic functions, e.g. Garnett, *Bounded Analytic Functions*, Theorem II.6.1 together with the exponential-type-0 hypothesis forcing the product to be finite). This chain of reasoning is not given in the text.

What would establish it: Provide a non-circular argument: state that B is entire of exponential type 0 (since $h_B = h_{Y_+} - h_O = 0$ by the indicator-subtraction property for products), with $|B(u)| = 1$ on $\mathbb{R}$; then cite the theorem that an entire function f of exponential type satisfying $|f| = 1$ on $\mathbb{R}$ and $|f| \leq 1$ on $\mathbb{C}^+$ is a finite Blaschke product (a consequence of the Hadamard factorization of entire functions of order 1 with $|f| = 1$ on $\mathbb{R}$ and type 0, whose order-1 entire Hadamard factors with purely imaginary zeros cancel by the boundary condition, leaving only finitely many upper

half-plane zeros contributing a finite Blaschke product and finitely many real zeros contributing real linear factors, and the product of real linear factors with modulus 1 on $\mathbb{R}$ must be empty).

Finding 7 — Not fixable as stated

Location: Main Structural Theorem, Step 6, sentences “The factorization $Y_+ = B \cdot O \dots$ translates, under the Paley–Wiener isomorphism $L^2[0, 1] \cong PW_1 \cap H^2(\mathbb{C}^+)$, to a factorization of $d\xi$ as a convolution on $[0, 1]$: $d\xi = \hat{B} * dO$ ” and “Convolution with $\hat{B} \neq \delta_0 \dots$ introduces correlations between increments of $d\xi$ at different frequencies, contradicting the Cramér orthogonality”.

Claim as written: The pointwise factorization $Y_+(z) = B(z) \cdot O(z)$ in $\mathbb{C}^+$ translates to a convolution $d\xi = \hat{B} * dO$ of random spectral measures on $[0, 1]$, and this convolution introduces correlations in $d\xi$, contradicting the Cramér orthogonality of $d\xi$.

What is relied on but not established in the text: This is the crux argument of the paper, and it is not established. Two independent gaps are present.

Gap (a): The convolution identity $d\xi = \hat{B} * dO$ is not established.

The Paley–Wiener isomorphism $L^2[0, 1] \cong PW_1 \cap H^2(\mathbb{C}^+)$ is a Hilbert space isomorphism between deterministic L^2 functions and deterministic entire functions in the Paley–Wiener class. The Cramér spectral measure $d\xi$ is a *random* measure on $[0, 1]$: for each $f \in L^2[0, 1]$ the stochastic integral $\int_0^1 f(\mu) d\xi(\mu)$ is a mean-square-defined random variable. The product $Y_+(z) = B(z)O(z)$ is a *deterministic* identity of entire functions (if B is a fixed, deterministic Blaschke product and O is the deterministic outer factor of Y_+). Under the Paley–Wiener isomorphism, this identity says $\hat{Y}_+ = (\hat{B}) \cdot (\hat{O})$ as L^2 functions (here $\hat{\cdot}$ denotes Fourier transform as a Paley–Wiener element), but this is a convolution of $L^2[0, 1]$ functions, not of random measures. The Cramér representation $Y_+(u) = 2 \int_0^1 e^{i\mu u} d\xi(\mu)$ equates the deterministic function Y_+ with a stochastic integral. The text identifies the Fourier coefficient $\hat{Y}_+(\mu)$ with $d\xi(\mu)$, but $d\xi$ is a random orthogonal measure, not the Fourier transform of the deterministic function Y_+ . The equation $d\xi = \hat{B} * dO$ (where dO is described as “the spectral representation of O ”) conflates the deterministic Fourier coefficient of O with a random spectral increment of a process whose covariance spectral density is $|O|^2 d\mu$. To make sense of $d\xi = \hat{B} * dO$ as a random measure identity, one would need to define dO as a separate random measure and explain how it relates to $d\xi$; the text does not do this.

Gap (b): Convolution does not in general destroy the orthogonal-increment property.

Even granting a formal interpretation of $d\xi = \hat{B} * dO$, the claim that “convolution with $\hat{B} \neq \delta_0$ introduces correlations between increments of $d\xi$ at different frequencies” is not established. On the Fourier side, multiplication of two functions (i.e. pointwise product) of L^2 corresponds to convolution on the physical side. Here, $Y_+(z) = B(z)O(z)$ in frequency space (i.e. the Fourier transform of Y_+ is the convolution of $\hat{B}$ and $\hat{O}$), so the *spectral density* of Y_+ is $|\hat{Y}_+|^2 = |\hat{B}|^2 * |\hat{O}|^2$ (a convolution of spectral densities only if $\hat{B}$ and $\hat{O}$ have independent phases, which is not the case in general). For the *spectral measure* ρ of Y , we have $d\rho(\mu) = |Y_+(\mu)|^2 d\mu = |B(\mu)|^2 |O(\mu)|^2 d\mu$ (since $|B| = 1$ on $\mathbb{R}$, this equals $|O(\mu)|^2 d\mu$). Therefore the spectral density of Y equals $|O|^2$ regardless of whether B is trivial or not, and the Cramér orthogonal increment measure $d\xi$ satisfies $\mathbb{E}[|d\xi(\mu)|^2] = d\rho(\mu) = |O(\mu)|^2 d\mu$ in both cases $B \equiv 1$ and $B \not\equiv 1$. The Cramér orthogonality $\mathbb{E}[d\xi(\mu)d\xi(\mu')] = 0$ for $\mu \neq \mu'$ is a property of the *process* Y , not of the analytic function Y_+ : $d\xi$ is orthogonal because Y is (assumed to be) a stationary process with spectral representation, and the orthogonality is automatic for any weakly stationary process, irrespective of the Blaschke factor. A non-trivial B would change the zeros of Y_+ in $\mathbb{C}^+$ but would not alter $|Y_+(u)| = |O(u)|$ on $\mathbb{R}$ (since $|B| = 1$ on $\mathbb{R}$), and hence would

not affect the spectral density $d\rho = |Y_+|^2 d\mu = |O|^2 d\mu$ of the process. Therefore the convolution $d\xi = \hat{B} * dO$, even if formally well-defined, does *not* introduce correlations in the spectral measure of Y : the covariance kernel $R(\tau) = \int e^{i\mu\tau} d\rho(\mu)$ depends only on $d\rho = |Y_+|^2 d\mu$, which is unchanged by a unimodular factor B on $\mathbb{R}$. The text's claim that a nontrivial Blaschke factor produces a contradiction with Cramér orthogonality is therefore not established by the given argument.

What would establish it: A correct argument closing Step 6 would require a different route. One possible route: show directly that $B \equiv 1$ by exploiting the one-sidedness of the Fourier support of Y_+ together with a rigidity theorem for functions in the Hermite–Biehler class (e.g. de Branges's theorem that a function in the HB class with real zeros is a product of a real entire function and an exponential, but this does not immediately apply). Another possible route: use the fact that B is entire and of exponential type 0 with $|B| = 1$ on $\mathbb{R}$ and $|B| \leq 1$ on $\mathbb{C}^+$ to conclude directly that B is a finite Blaschke product (as in Finding 6), and then show that a finite Blaschke product with zeros in $\mathbb{C}^+$ cannot be entire unless the product is trivial (which is false for finite products — a finite Blaschke product is meromorphic with poles at the reflected zeros $\bar{z}_n$ in $\mathbb{C}^-$, but if both Y_+ and O are entire and $B = Y_+/O$, then B has no poles, so O must vanish at each $\bar{z}_n$ — but O is outer and nonzero in $\mathbb{C}^+$, and zeros of O in $\mathbb{C}^-$ are possible; this path requires a separate argument). The document does not provide any of these alternatives.

All propositions in the document other than those addressed in Findings 1–7 are established by the document as written or by standard results whose hypotheses are clearly satisfied: the band-limitedness conclusion from `BandLimitedness.tex` ($\text{supp } \rho \subseteq [-1, 1]$), the Cramér spectral representation, the definition and entire-function property of Y_+ , Lemma (Indicator of Y_+), the inner-outer factorization in Step 1, and the Theorem (Reality) as a consequence of the HB reduction and the Main Structural Theorem are all within the scope of what the document establishes subject to the gaps in Findings 1–7. Theorem B of `three_missing_theorems.pdf` (the diagonal-dominance identity for the whitened differential) is not invoked in `RealityOfZeros.tex` and no gap arising from its absence was found.